

Hee Won Lee

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SUMMARY

Cloud infrastructure and software engineering expert with deep experience in designing, implementing, and optimizing distributed systems across cloud-native, hybrid, and on-premises environments. A rapid adopter of emerging technologies, with specialized expertise in infrastructure as code, Kubernetes orchestration, observability, cloud security, and AI-integrated systems. At Samsung Electronics, led a team of Cloud Architects and Software Engineers to architect and automate compute, storage, and networking infrastructure across AWS, Azure, and on-premises datacenters—delivering large-scale automation, telemetry, and intelligent platform services. At AT&T Labs, led the development of Kubernetes-based zero-touch provisioning and delivered performance optimization for containerized storage and in-memory systems. Holds a Ph.D. in Computer Science with a focus on virtualization (QEMU-KVM), cloud networking, and systems performance. Certified Microsoft Azure Solutions Architect Expert and Cybersecurity Architect Expert.

EXPERIENCE

Samsung Electronics, Hwaseong, South Korea

- Principal Engineer Dec 2020 – Feb 2024
 - Infrastructure Automation
 - Designed and implemented auto-provisioning of AWS/Azure resources using Terraform and Bicep
 - Built a GitHub Actions pipeline to deploy web apps to Azure App Services
 - Developed a monitoring and alerting system for Kubernetes clusters using the Prometheus Stack (Prometheus, Grafana, and Alertmanager)
 - Developed a real-time monitoring and telemetry system for VMs using Telegraf, InfluxDB, and Grafana
 - AI Integration
 - Integrated Azure AI Search into an on-premises chat service built on LangChain, leveraging a large language model with retrieval-augmented generation (RAG).
 - Led prompt engineering strategies to optimize LLM responses and system integration.
 - Cloud Security with Zero Trust Principles
 - Enforced Multi-Factor Authentication (MFA) based on user location and sign-in risk
 - Assigned just-in-time (JIT) administrator roles with maximum allowable durations
 - Created Microsoft Sentinel detection rules using KQL and monitored threats using the MITRE ATT&CK framework
 - Leadership & Architecture
 - As Head of the Cloud Platform Group, led team growth from 30 to over 50 engineers in three years by actively recruiting Software Engineers, DevOps Engineers, and Cloud Architects.
 - Designed and managed hybrid cloud infrastructure across AWS, Azure, and on-premises datacenters, supporting 10,000+ VMs, 10,000+ VDIs, and Active Directory services for 60,000+ user accounts.

AT&T Labs Research, Bedminster, New Jersey, USA

- Principal Inventive Scientist Mar 2015 – Jul 2020
 - Zero Touch Provisioning (based on Kubernetes)
 - Designed and implemented auto-provisioning for Kubernetes clusters using Kubeadm.
 - Designed and implemented containerized Ceph storage with Kubernetes helm charts
 - Performed resiliency tests for containerized Ceph storage with Kubernetes
 - Developed Redis High Availability with Kubernetes in a CI/CD pipeline (based on Gerrit and Jenkins)
 - Integrated Redis HA helm charts into Radio Access Network Intelligent Controller
 - Designed and implemented a system that controls network bandwidth/delay between multi-site OpenStack clusters (using Ansible)
 - System Performance Engineering
 - Evaluated the performance of open-source object storage systems (Ceph vs. Swift)
 - Evaluated the performance of page cache & dm-cache in virtualized environments
 - Evaluated the performance of Redis with Intel Optane Persistent Memory
 - Evaluated and compared the performance of *Redis* vs *Aerospike* vs *Cassandra* using the YCSB benchmark tool
 - Set up a testbed for NVMe over Fabrics with CPU offloading on RoCE-based lossless networks, and then evaluated the performance and verified zero CPU usage on the target host machine
- Internship May 2014 – Aug 2014
 - Designed and implemented I/O traffic protection mechanisms for cloud storage

North Carolina State University, Raleigh, North Carolina, USA

- Research Assistant Jan 2010 – Dec 2014

- Designed and implemented a virtualization platform for evaluating distributed applications running on diverse OSs (Linux, FreeBSD, Windows, JunOS)
- Redesigned and customized the QEMU-KVM hypervisor to synchronize distributed VMs with virtual time

Korea Telecom, Daejeon, South Korea

- **Software Engineer** Jan 2002 – Jul 2009
 - Capacity Planning for KT Backbone Network and IPTV Services
 - Built a virtual network environment of KT backbone networks using OPNET simulation tool
 - Performed link capacity planning for IPTV services
 - VoIP Network Management System
 - Conducted requirement management and then designed architecture of VoIP-NMS
 - Deployed VoIP-NMS to production
 - IP Core Network Management System
 - Designed and implemented a system component that collects network equipment information from MIB (Management Information Base) using SNMP (Small Network Management Protocol)
 - Developed SQL scripts for Oracle database
 - Developed a system component that exchanges information with Service Assurance System using XML-RPC
 - Set up an MPLS testbed using Cisco Routers 7204/7500, Juniper Router M5 and Ixia Traffic Generator

CERTIFICATIONS

- **Microsoft Azure Solutions Architect Expert**
 - Microsoft Certified: Azure Administrator Associate (AZ-104)
 - Microsoft Certified: Azure Solutions Architect Expert (AZ-305)
- **Microsoft Cybersecurity Architect Expert**
 - Microsoft Certified: Azure Security Engineer Associate (AZ-500)
 - Microsoft Certified: Cybersecurity Architect Expert (SC-100)

EDUCATION

North Carolina State University, Raleigh, North Carolina, USA

- **Ph.D. in Computer Science** Aug 2009 – May 2015
 - Thesis: Network Emulation with Adaptive Time Dilation
 - Adviser: Prof. Mihail L. Sichitiu, Co-adviser: Prof. David Thuente
 - Concentration: Virtualization, Networked Systems, Distributed Systems

Carnegie Mellon University, Pittsburgh, Pennsylvania, USA

- **M.E. in Software Engineering** Aug 2004 – Aug 2005

Korea University, Seoul, South Korea

- **B.E. in Electrical Engineering** Mar 1995 – Feb 2002

SOFTWARE DEVELOPMENT EXPERIENCES

Virtualization

- **Hypervisor (QEMU-KVM) [C++]**
 - Designed and implemented a spinlock on shared memory to synchronize distributed VMs (processes created by QEMU-KVM hypervisor) and simulation nodes (processes created by NS-3) with virtual time
 - Implemented a virtual time synchronization daemon with TCP/UDP sockets using Boost Asio Library
- **High-Performance Emulation System [C++, Python]**
 - Emulated higher performance with a time dilation technique using unmodified OSs (Linux, FreeBSD, Windows, and Junos) and real application workloads (VLC media player)
 - Modified the QEMU-KVM hypervisor for virtual time
- **Hybrid Simulation and Emulation System [C++, Python]**
 - Built an integrated simulation (NS-3) and emulation (QEMU-KVM) framework in a distributed environment

Storage

- **Cloud Storage**
 - Designed a system architecture for making a commodity SSD array a usable iSCSI block storage backend for OpenStack clusters
 - Designed and implemented the backend of OpenStack Cinder for dynamic configuration and QoS support
 - Designed and implemented containerized Ceph with Kubernetes helm charts with an one-osd-per-pod approach
 - Evaluated the performance of Ceph with host/vm page cache, dm-cache, and rbd cache
 - Evaluated and compared the performance of SATA SSD and NVMe SSD
- **Persistent Memory**
 - Set up a testbed with Intel Optane Persistent Memory (in collaboration with Intel team)
 - Analyzed the performance characteristics of Intel Optane Persistent Memory

- Measured the latency of Redis with Intel Optane Persistent Memory using the *memtier_benchmark* tool
- I/O Traffic Protection [Python]
 - Designed and implemented an I/O bandwidth reservation algorithm that guarantees the minimum I/O bandwidth of a virtual host connected to a shared storage volume through iSCSI
 - Designed a NUMA-aware CPU pinning algorithm that protects I/O traffic from CPU interference
 - Evaluated the system performance for diverse types of I/O pattern using FIO and Filebench
- RDMA over Converged Ethernet (RoCE)
 - Set up a testbed for lossless RoCE using PFC-capable Mellanox Spectrum Switch and Dell servers equipped with ConnectX-5 NICs
 - Evaluated the performance of NVMe over Fabric for zero CPU usage on target host machine

Networking

- Network Traffic Controller [Ansible]
 - Designed and implemented a system that controls network bandwidth/delay between OpenStack clusters using Tc/NetEm
- Network Link Emulation [C++, Python]
 - Interconnected distributed VMs through virtual links using Tc/NetEm with adaptive time dilation
- Software-Defined Networking [Python]
 - Used OpenFlow POX controller and Open vSwitch to migrate VirtualBox-based virtual networks
- Routing Algorithms [Java]
 - Implemented modified Dijkstra's algorithm with negative weights
 - Implemented vertex/edge disjoint paths and elementary circuits search algorithms
 - Implemented CPLEX code with Integer Linear Programming for Routing and Wavelength Assignment
- Queueing Systems [Java]
 - Implemented discrete event simulation models for queueing systems (M/M/1, M/M/m, M/G/m)
- Wireless Networking [C++]
 - Designed and implemented an OMNeT++ model for IVG (Inter-Vehicle Geocast) routing protocol in vehicle-to-vehicle ad-hoc network simulation
 - Connected KVM-based virtual nodes through a wireless simulation model of OMNeT++
- Peer-to-Peer File Sharing System [Java]
 - Mapped a file signature (created by encoding text document with m-bit string) onto a hash space
 - Used Apache Mina, Apache FtpServer, and JavaDB

CPU & Memory

- CPU/Memory Control [Python]
 - Designed and implemented a CPU/Memory control algorithm that enforces CPU/Memory ceiling and allocates a relative share of CPU time using Linux Control Groups (Cgroups)
- Multi-Threading [C++]
 - Created a thread for searching text file paths and multiple worker threads for counting words using Boost Filesystem and Thread Libraries
 - Implemented multiple producer, multiple consumer thread-safe queue and map
- Prioritized Preemptive Schedulers [C]
 - Implemented a prioritized preemptive scheduler using POSIX thread library
 - Designed and implemented a priority-based scheduling algorithm using XINU kernel (a small UNIX OS)
- Demand Paging [C]
 - Implemented a demand paging system that allows for more address space than physically available one (XINU)

PUBLICATIONS

CONFERENCES

- [9] **Dejavu: Reinforcement Learning-based Cloud Scheduling with Demo and Competition.**
Seonwoo Kim, Yoonsung Nam, Minwoo Park, Hee Won Lee, Seyeon Kim, Sangtae Ha.
In *IEEE 21st International Conference on Mobile Ad-Hoc and Smart Systems (MASS 2024)*.
Seoul, South Korea, Sep 2024.
- [8] **TIPS: Making Volatile Index Structures Persistent with DRAM-NVMM Tiering.**
Madhava Krishnan, Wook-Hee Kim, Xinwei Fu, Sumit Kumar Monga, Hee Won Lee, Minsung Jang, Ajit Mathew, and Changwoo Min.
In *USENIX Annual Technical Conference (ATC 2021)*.
Virtual Event, Jul 2021.

- [7] **eMRC: Efficient Miss Ratio Approximation for Multi-Tier Caching.**
Zhang Liu, Hee Won Lee, Yu Xiang, Dirk Grunwald, and Sangtae Ha.
In *USENIX 19th Conference on File and Storage Technologies (FAST 2021)*.
Virtual Event, Feb 2021.
- [6] **EF-Dedup: Enabling Collaborative Data Deduplication at the Network Edge.**
Shijing Li, Tian Lan, Bharath Balasubramanian, Moo-ryong Ra, Hee Won Lee, and Panta Rajesh.
In *IEEE 39th International Conference on Distributed Computing Systems (ICDCS 2019)*.
Dallas, TX, USA, Jul 2019.
- [5] **Fighting with Unknowns: Estimating the Performance of Scalable Distributed Storage Systems with Minimal Measurement Data.**
Moo-ryong Ra and Hee Won Lee.
In *IEEE 35th Symposium on Mass Storage Systems and Technologies (MSST 2019)*.
Santa Clara, CA, USA, May 2019.
- [4] **MIST: Mitigating Host-Side Interference for Storage Traffic in Virtualized Data Centers.**
Hee Won Lee and Moo-ryong Ra
In *IEEE 9th International Conference on Cloud Computing (CLOUD 2016)*.
San Francisco, CA, USA, Jun 2016.
- [3] **Integrated Simulation and Emulation using Adaptive Time Dilation.**
Hee Won Lee, David Thuente, and Mihail L. Sichitiu.
In *ACM 2nd SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS 2014)*.
Santa Clara, CA, USA, May 2014.
- [2] **Resource Management System for Next Generation Services.**
Seung-Hee Han, Bom-Su Kim, Chan-Kyou Hwang, Hee Won Lee, Byung-deok Chung.
In *IEEE International Conference on Advanced Technologies for Communications 2019*.
Hai Phong, Vietnam, Oct 2009.
- [1] **Reusability Enhancement by Using Flexible Topology Architecture for Network Management System.**
Hee Won Lee, Chan Kyou Hwang, Jae-Hyoung Yoo, Ho-Jin Choi, Sungwon Kang, and Dan H. Lee.
In *IEEE/ACIS 7th International Conference on Computer and Information Science (ICIS 2008)*.
Portland, Oregon, USA, May 2008.

JOURNALS

- [3] **FPV Video Adaptation for UAV Collision Avoidance.**
Simran Singh, Hee Won Lee, Tuyen X. Tran, Yu Zhou, Mihail L. Sichitiu, Ismail Güvenç, and Arupjyoti Bhuyan.
IEEE Open Journal of the Communications Society, vol. 2, pp. 2095–2110.
Aug 2021.
- [2] **Network Link Emulation With Adaptive Time Dilation.**
Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
Journal of Parallel and Distributed Computing, vol. 104, pp. 88–98.
Jun 2017.
- [1] **High-performance Emulation of Heterogeneous Systems using Adaptive Time Dilation.**
Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
International Journal of High Performance Computing Applications, vol. 29, issue 2.
May 2015.

WORKSHOPS, POSTERS AND TECHNICAL REPORTS

- [3] **Accelerating Applications in the Fast-Moving Devices with Proactive Provisioning (Poster).**
HyunJong Lee, Hee Won Lee, Moo-Ryong Ra, Yu Xiang, and Jason Flinn.
In *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys 2019)*.
Seoul, South Korea, Jun 2019.

- [2] **IOArbiter: Dynamic Provisioning of Backend Block Storage in the Cloud.**
Moo-Ryong Ra and Hee Won Lee.
arXiv preprint arXiv:1904.09984.
Apr 2019.
- [1] **End-User IPTV Traffic Measurement of Residential Broadband Access Networks.**
Young J. Won, Mi-Jung Choi, Byung-Chul Park, James W. Hong, Hee Won Lee, Chan Kyu Hwang,
and Jae-Hyoung Yoo.
In *IEEE Network Operations and Management Symposium (NOMS) Workshops*.
Apr 2008.

PATENTS

- [1] **Dynamic Provisioning of Storage in the Cloud.**
Moo-ryong Ra and Hee Won Lee.
US 10,530,703 B2.
Date of Patent: January 7, 2020.

RESEARCH MENTORING

- **Simran Singh**, North Carolina State University
 - Project: UAV real-time video applications with cellular infrastructure support
 - 2019 fall internship at AT&T
- **Madhava Krishnan**, Virginia Tech
 - Project: A general purpose in-memory key-value store architecture for emerging persistent memory
 - 2019 summer internship at AT&T
- **HyunJong (Joseph) Lee**, University of Michigan–Ann Arbor
 - Project: Predictive migration of edge computations
 - 2018 summer internship at AT&T and 2021 winter internship at Samsung
- **Zhang Liu**, University of Colorado–Boulder
 - Project: Multi-tier cache orchestration with miss ratio curve
 - 2017 summer internship at AT&T

HONORS & AWARDS

- **Outstanding Teaching Assistant**, North Carolina State University Apr 2011
 - Awarded the day of 29 April 2011
 - For excellent service as a teaching assistant during the 2010-11 calender year, the Department of Computer Science recognizes Hee Won Lee as an Outstanding Teaching Assistant
 - Teaching Assistant Courses
 - Algorithm (2011)
 - Operating System (2010, 2012, 2013)
 - Computer Networks (2011, 2012)
 - Software Engineering (2010)
- **Army Commendation Medal**, U.S. Army Feb 2001
 - Awarded an ARCOM (Army Commendation Medal) for dedications and outstanding leadership in U.S. Army

REFERENCES

- **Prof. Mihail L. Sichitiu**
(PhD Program Advisor)
Deaprtment of Electrical and Computer Engineering, North Carolina State University
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- **Prof. Sangtae Ha**
(Research Collaborator)
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- **Dr. Yu Xiang**
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